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ON

AUTOCTI : Automated Cyber Threat Detection using AutoML

Submitted in partial fulfillment for the award of degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND DESIGN

Submitted by

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CANARA ENGINEERING COLLEGE
(An Autonomous Institution, Under VTU, Belagavi and Recognized by AICTE, Accredited by NBA (CSE, ISE, ECE) and NAAC 'A' GRADE)
Sudhindra Nagar, Benjanapadavu, Mangaluru - 574219, Karnataka.

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the bonafide students of VII semester COMPUTER SCIENCE AND DESIGN in partial fulfillment for the award of Bachelor of Engineering in COMPUTER SCIENCE AND DESIGN of the Visvesvaraya Technological University, Belagavi, during the year 2025-2026. This is to certify that all the corrections and suggestions indicated during the Internal Assessment have been duly incorporated. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

We hereby declare that our entire work embodied in this Project Report titled **"AUTOCTI : Automated Cyber Threat Detection using AutoML"** has been carried out at CANARA ENGINEERING COLLEGE, Mangaluru, under the supervision of **Mr. Ravishankara Kulamarva**, for the award of **Bachelor of Engineering** in **COMPUTER SCIENCE AND DESIGN**. This report has not been submitted to any other University for the award of any other degree.

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Abstract

In the fast-forwarded world of today, every organization seems to be at risk from cyber threats and detecting those cyber threats has become more and more complex. Traditional manual method of collecting threat data scraping from dark web forums, security blogs is time-consuming and mostly fails. These provide alerts on the threat. There could be delays, which could be extremely dangerous. The creation of an automated cyber threat detection system using AutoML is the aim of this project. It aims to automate all the steps starting from data collection, classification and response. This system gathers threat information from various internet sources and categorizes it based on the types and severity of the threats, facilitating security teams prioritization of which one to investigate first. A dashboard driven by AI will offer real-time updates on threats that have been identified. The key feature of this system is that it has an auto-delete feature, which automatically deletes or isolates the high-severity threat before it can cause any harm occur. This feature reduces manual efforts and increases the cyber security. Additionally, it can be improved and expanded upon in the future.

Keywords : Cyber threat detection, AutoML, Classification based on severity, AI Powered dashboard, auto-delete/block

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Chapter 1

Introduction

1.1 Background

Traditional monitoring of dark web forums, security blogs, and other platforms is performed manually by cybersecurity analysts. This method is more time-consuming and prone to mistakes[1]. To overcome these challenges, we introduce a project which will automatically detect the cyber threat by using AutoML and at the same time delete the detected threat automatically[2]. This project also provides an AI-powered dashboard, which gives real-time insight and alerts to the security team[2].

1.2 Motivation and Problem statement

Problem Statement: To develop an AutoML powered system to analyze the threat data and improve detection, addressing the limits of manual cyber threat analysis using AutoML.
Motivation: Major concerns of the growing volume of cyber threats among organizations are due to the fact that traditional methods of threat detection remain highly manual and time-consuming, so much that security teams cannot manage the disorganized flow in threat detection. This is leading to an increase in response time and a vulnerability to breaches[3]. Recent development in the field of AutoML introduces an opportunity for the automation of cyber-threat detection and classification[2].

1.3 Objectives

- To build a system using AutoML that can automatically identify, classify and delete/block high-risk cyber threats based on their type and severity.
- To create an easy to use dashboard that shows useful information about detected threats in real-time, helping security teams quickly understand, respond to, or take action.

1.4 Scope and limitations

This project aims to build an automated cyber threat detection system that can find and classify threat data from different sources. AutoML will help this system identify and categorize threat data by severity. The system will then give real-time alerts for unknown threats on an AI-powered dashboard[2]. It will help organizations recognize and respond to cyber threats as they happen[2].

1.5 Relevance and Type

In today's digital world, cyber threats are becoming common[1]. The struggle to keep up with these attacks is real. Currently, detecting these threats is done manually, which takes a lot of time and is also prone to mistakes[3]. This project aims to solve the problem by automatically detecting cyber threats and either deleting or isolating them based on their severity[2]. We use AutoML-powered machine learning models to monitor network traffic and identify security breaches.[4] This project combines machine learning, data analysis, and network security to offer practical solutions. With a complexity level that ranges from intermediate to advanced, it is well-suited for both academic research and field implementation, especially in enterprise and cloud environments.

1.6 Organization of the report

1.6.1 Organization of the Report

This project report is organized into seven major chapters as described below:

- **Chapter 1 – Introduction:** Provides the background of the study, motivation, objectives, scope, limitations, relevance, and type of the project undertaken.
- **Chapter 2 – Literature Survey:** Reviews the existing research articles related to cyber threat detection using AI and highlights the gaps and findings, followed by a comparison and summary.
- **Chapter 3 – Software Requirements Specification:** Defines the functional and non-functional requirements, user interface design, and hardware/software specifications essential for the proposed system.
- **Chapter 4 – System Design:** Describes the architectural, functional, and control flow design of the system with diagrams such as modular design, sequence diagram, and data flow diagrams.
- **Chapter 5 – Implementation:** Explains the technologies used, development modules, algorithms, and coding approaches implemented in the system.
- **Chapter 6 – Results and Discussion:** Presents the platform output, testing methods, evaluation criteria, comparative analysis, and system performance.
- **Chapter 7 – Conclusions and Future Work:** Summarizes the project outcomes, key contributions, and provides recommendations for future enhancements.

Chapter 2

Literature Survey

2.1 Article 1 :A Review on Darknet and Cyber Threat to Society, Counter Measures and Challenges and Trends in Cyber Security and Intrusion Detection

Authors: B Ganga, BTLata, TRPallavi, and KR Venugopal[1]

2.1.1 Brief Findings of Article 1

The paper reviews the increasing threat of cybercrime made possible by the darknet, which gives hackers the encryption and anonymity features they want. Based on the type of malware used, injection techniques, social engineering, and network intrusion, the review investigates various cyberthreats. The study describes how the darknet facilitates illicit activities like the sale of weapons and drugs, human trafficking, ransomware attacks, and data breaches. It then emphasizes how dynamic and constantly evolving today's cyberthreats have become, necessitating equally dynamic cyberdefense tactics. The study also identified several cybersecurity measures that safeguard a network and data, such as incident response planning, firewalls, intrusion detection systems, antivirus software, and staff training. AI, machine learning, and quantum computing are urgently needed to enhance intrusion detection systems so they can respond to threats that are not immediately apparent. The authors then

enumerate additional contemporary concerns, including policy gaps, social engineering, automation vulnerabilities, and cloud security.

2.1.2 Design/Methodology/Techniques Adopted in Article 1

The authors of this paper use a thorough literature review methodology to examine cybersecurity risks, darknet activity, and contemporary defence strategies. They have used keywords like "cybersecurity threats," "darknet," "cyber-attacks," and "intrusion detection" to search several trustworthy databases, including IEEE Xplore, Google Scholar, and ResearchGate, in order to find research publications from the previous fifteen years. Based on these, a variety of cyberthreats, such as malware-based threats, code injection attacks, social engineering techniques, active and passive threats, and network-based threats, have been identified and categorised into technical and behavioural characteristics.

The paper also explores the darknet in detail, explaining how Tor, I2P, FreeNet, and ZeroNet enable anonymised and encrypted communications, as well as how its infrastructure is designed to set it apart from the surface and deep web. This anonymity fosters a number of illegal activities, including ransomware-as-a-service, drug and human trafficking, and cyberespionage, which are far more difficult to identify and stop. The authors use well-known darknet sites like Silk Road, AlphaBay, and Hydra to show how widespread and unchecked darknet crimes are.

It describes a number of cybersecurity tactics, including response mechanisms like staff training and incident response planning, detection tools like malware scanners, audit programs, and intrusion detection systems, and preventive measures like encryption, strong password policies, and frequent software updates. In order to help people identify and react to social engineering attacks, it is especially important to raise awareness through education and simulation-based training. Tools for system monitoring and visualisation help administrators better understand and react to cyber incidents in real time. When combined, this methodical framework offers a thorough grasp of the state of cyberthreats today as well as the necessary

defences.

2.1.3 Results Achieved in Article 1

- **Detailed Threat Classification:** Cyber threats were categorized into groups based on networks, malware, code injection, and social engineering..
- **Darknet Crime Insights:** highlighted how illegal activities like drug trafficking, ransomware, and data breaches are made easier by the darknet.
- VBA achieved a **CMAE of 4.25**, outperforming:
  - **Layered Security Measures:** Technical tools, employee training, and incident response are recommended for enhanced cybersecurity.
- Future Tech Adoption placed a strong emphasis on the application of AI, ML, and quantum learning for advanced threat detection.
- Low user awareness, cloud vulnerabilities, and darknet anonymity are among the difficulties noted.

2.2 Article 2 : AI-Enhanced Cyber Security Proactive Threat Detection and Response Systems

Authors: Venkata Sai Swaroop Nallapareddy, et al[2]

2.2.1 Brief Findings of Article 2

The integration of artificial intelligence into contemporary cybersecurity is covered in the article "AI-Enhanced Cyber Security: Proactive Threat Detection and Response Systems."It talks about how AI can greatly improve threat detection and response through automation, real-time analytics, and ma chine learning.While traditional security measures are slow

and inaccurate, artificial intelligence (AI) can process massive datasets, spot anomalies, and accurately detect zero-day attacks.layers for supervised and unsupervised real-time threat detection, automated incident response, and behavior analytics for the suggested system consists of detection.Additionally, it emphasizes ongoing education and smooth integration with the current security framework. It operates through techniques that prevent information leaks.

2.2.2 Design/Methodology/Techniques Adopted in Article 2

Using the multi-layered design methodology, the research suggests an AI-driven cybersecurity system that offers automated response, adaptive learning, and real-time threat detection. The system uses a variety of artificial intelligence techniques, such as behavior analytics, machine learning, and natural language processing. Supervised models are trained on historical attack data to identify known threats, whereas unsupervised models use patterns in network traffic and user behavior to find anomalies and new, unidentified threats. By examining the textual communication patterns between emails and social media platforms, natural language processing (NLP) is particularly useful in spotting phishing and social engineering attempts. Predictive analytics finds patterns that could trigger future attacks and finds vulnerabilities before they can be used against an organization. Through a continuous learning loop that incorporates human expert feedback and reinforcement learning, people adjust to new threats. Large data environments are supported by cloud scalability, and integration with current SIEM and threat intelligence systems is made possible by the architecture. Additionally, sensitive data can be protected during the training and operation of AI models using privacy-preserving technologies like federated learning and differential privacy.

2.2.3 Results Achieved in Article 2

- **Faster Reaction to Serious Threats:** By responding quickly to significant threats, the response mechanism minimized potential harm

and decreased the network's susceptibility to cyberattacks.

- **High Scalability in Cloud Environments:**The system showed its ability to operate dependably in extensive cloud-integrated infrastructures by maintaining its momentum as the number of connected devices and data size increased.
- **Decrease in False Positives:** As AI models developed over time, the number of false alarms decreased, freeing up security professionals to focus on actual threats and cut down on needless work.

2.3 Article 3 : Cyber Threat Hunting to Detect Unknown Threats in the Enterprise Network

Authors: Arssy Hasyir Nursidiq, et al[3]

2.3.1 Brief Findings of Article 3

To help, identify unknown threats, the study proposed a proactive cyber threat hunting model designed for enterprise networks. By using packet capture, long-tail analysis, and similar incident detection methods with the MITRE ATTACK framework and Diamond Model, the researchers discovered malicious documents in the form of .html and .js files. They also found suspicious IP addresses that bypassed standard security systems. The model aimed to improve future protection by comparing results against security devices and using disk forensics to confirm the threats. The findings show that proactive threat hunting improves situational awareness, provides early warnings, and strengthens organizational defenses. This approach also leads to more automation and speeds up the identification of threats and responses through machine learning.

2.3.2 Design/Methodology/Techniques Adopted in Article 3

The three main phases of the active threat-hunting process in the research focus on enterprise networks: Collection, Analysis and Validation, and De-

ployment. In the collection phase, packet capture tools track and record network traffic to identify irregularities. During the analysis and validation phase, long-tail analysis finds outliers. A hypothesis about threats is formed and confirmed using the MITRE ATTCK framework, Cyber Kill Chain, and Diamond Model. Forensic software, like FTK Imager and Autopsy, is used to analyze disks to confirm malicious activities. The results are then turned into use cases for the deployment stage. This improves detection abilities and updates security device rules. This approach ensures effective cyber defense by combining threat intelligence with manual investigation.

2.3.3 Results Achieved in Article 3

- **Identification of Undetected threats:** The system found malicious.html and malicious.js files that other security systems did not recognize. It also detected harmful traffic from more than 76,000 IP addresses.
- **Accurately Validating Threats:** The forensic analysis was reliable. Different files, such as Trojan: Win32/Leonem, were confirmed to be harmful using tools like VirusTotal and Windows Defender logs.
- **Increased Response Measures:** Security rules, such as blocking harmful IPs to prevent future attacks, were improved by using threat intelligence.
- **Enhanced Threat Visibility:** To provide a clear view of attacker activity and possible weaknesses, the study combined the Diamond Model with the MITRE ATTACK framework.

2.4 Article 4: Web Data Mining for Cyber Security Threat Detection

Authors: Tejal Bhimraj Ghuge, Sammip Sunil Biradar.[5]

2.4.1 Brief Findings of Article 4

The use of web data mining to identify cybersecurity risks is covered in this article. Cyber threats have increased dramatically as a result of the growing use of the internet, impacting people, businesses, and governments. The study clearly shows how important machine learning, intrusion detection systems, and association rule mining are for identifying such threats and anticipating vulnerabilities. Twitter and JPMorgan are two examples of businesses that use data mining to better protect consumer information. The article's conclusion mentions how integrating Web Mining with AI would help detect malware and insider attacks, as well as identify malicious behaviour and secure networks.

2.4.2 Design, Methodology, Techniques Adopted in Article 4

The article explores data mining techniques in the context of cybersecurity threat detection using a thorough and methodical approach. Regression analysis, association rule mining, and classification are its three main approaches. Large amounts of data are automatically sorted into predefined classes by classification, which improves scalability and minimises the need for human intervention. However, accuracy is largely dependent on large amounts of data. Regression analysis is used to forecast potential threats in the future by taking into account historical data. However, incomplete or missing data—a problem that is inherent in cybersecurity environments—may limit its efficacy. Association rule mining turns out to be the most suitable method out of the three. Understanding attacker behaviour and identifying anomalies that might point to security breaches are two areas in which this discovery of hidden relationships and recurring patterns within datasets is especially helpful.

It then presents a clear data mining process flow. This flow starts with identifying reliable data sources, followed by careful cleaning to remove inaccuracies. It moves on to integrating data from various sources, selecting relevant information, and using suitable algorithms to uncover meaningful patterns. The paper also discusses how to improve threat de-

tection by using specialized cybersecurity tools. These include Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Endpoint Detection and Response (EDR), Network Detection and Response (NDR), and User and Entity Behavior Analytics (UEBA). Furthermore, the paper emphasizes that handling data should adhere to ethical practices and be guided by internationally recognized standards like ISO/IEC 27001 and legal frameworks such as the General Data Protection Regulation. This layered approach shows that thoughtful data mining can significantly improve an organization's cybersecurity defenses.

2.4.3 Results Achieved in Article 4

- **Threat Prediction:** Association rule mining proved most effective in detecting frequent anomalies and attacker behavior.
- **Automation of Detection:** Classification helped reduce manual work by automating data categorization
- **Security Enhancement:** Regression helped in predicting and mitigating future cyber vulnerabilities
- **Challenge Identification:** Key challenges include high computational cost, missing data, and the complexity of deep learning in real-time analysis.
- **Strategic Recommendations:** Use of recommender systems, AI-based automation, and adherence to legal-ethical norms like GDPR.
- **Future Scope:** Emphasis on enhancing big data handling and leveraging domain knowledge for more efficient mining and security.

2.5 Article 5 : The Performance Analysis of Cyber Threat Detection Model Using AI-based Classifiers

Authors: Navya Krishna Alapati, S.Dhanasekaran[6]

2.5.1 Brief Findings of Article 5

This paper looks at how well different AI-based classifiers, including Support Vector Machines, Random Forests, and Neural Networks, detect cyber threats. It shows that AI improves threat detection by providing high accuracy, low false positives, the ability to adjust to new threats, and quicker response times. The research emphasizes that traditional detection methods, like antivirus programs and firewalls, are no longer effective. AI-based solutions offer scalable, data-driven, and continuously evolving defense systems. The paper concludes that using these intelligent classifiers greatly improves the speed, accuracy, and reliability of cyber threat detection systems.

2.5.2 Design, Methodology, Techniques Adopted in Article 5

In this work, the authors presented a strong performance evaluation framework to identify cyber threats using different AI-based classifiers. The process starts with data preparation. They curated a large and diverse set of labeled cyber threat data, clearly distinguishing between “threat” and “non-threat” for model training. Next, they moved on to feature extraction. This phase identifies important indicators related to IP addresses, URLs, and common keywords that are used in cyber attacks. These features serve as the main inputs for the classifiers.

The training and model selection phase involves using various AI classifiers like Support Vector Machines, Random Forests, Decision Trees, and Neural Networks to assess their ability to distinguish between malicious and benign activities. In addition, the research also included deep learning models, such as LSTM and CNN, to explore how well they perform in real-time threat detection situations.

Performance metrics like accuracy, precision, recall, sensitivity, specificity, F1 score, and miss rate have been used to evaluate model effectiveness. The study also outlines the design of the detection model, explaining how AI systems analyze data streams to identify evolving anomalies or patterns that indicate cyber threats. This method highlights how adaptable

and scalable AI can be in cybersecurity.

2.5.3 Results Achieved in Article 5

- **Future Scope:** Emphasis on enhancing big data handling and leveraging domain knowledge for more efficient mining and security.
- **Scalability:** AI-based models adapt and learn from new attack patterns, making them more robust over time.
- **Reduced False Alarms:** Achieved a low miss rate and high precision, which helps reduce unnecessary alerts and optimizes response efforts
- **Improved Sensitivity Specificity:** Demonstrated strong ability to correctly detect actual threats while minimizing misclassification of benign activities.
- **Human-AI Collaboration:** The paper emphasizes that AI is a powerful augmentation tool, but should work alongside human analysts for best results.

2.6 Comparision Table

Table 2.1: Comparison of existing work and Gap Identification

Project Title and Author	Problem Addressed	Implementation and Results	Limitations/ Future Scope
Cyber Threat Hunting to Detect Unknown Threats in the Enterprise Network By Arssy Hasyir Nursidiq, et al	Existing security tools could not find advanced or unknown threats that were hidden in regular traffic. This caused weak situational awareness and delayed responses.	The model used packet capture,long-tail analysis, and forensic validation with MITRE ATTACK and the Diamond Model to find hidden threats. This led to successfully identifying malicious files and thousands of harmful IP addresses that traditional security tools overlooked.	Manual analysis is still needed. Future work will focus on using machine learning to detect threats more quickly and accurately.
AI-Enhanced Cyber Security Proactive Threat Detection and Response Systems By N.Sreenivasa et.al,2022	Traditional cybersecurity systems are slow and generate a lot of false alerts. They also have trouble identifying zero-day attacks. This makes proactive defense hard.	The AI-powered multi-layered system uses machine learning, NLP, and adaptive learning to identify and react to threats in real time. This leads to quicker responses, fewer false positives, and improved performance in cloud environments.	The system needs ongoing training and a solid AI setup. In the future, it will involve better privacy protection models and more automation for quicker and more independent threat responses.
Darknet and Cyber Threat to Society, Counter Measures and Challenges and Trends in Cyber Security and Intrusion Detection By B Ganga, BTLata	The increase in darknet-related cybercrimes and changing attack methods make traditional security measures inadequate for protecting networks and data.	The study identified darknet-enabled crimes, classified contemporary cyberthreats, and suggested layered defences using AI, ML, and emerging technologies for better threat detection.	The study calls for stricter regulations and AI-based security, highlighting low user awareness, cloud risks, and darknet anonymity.

Project Title and Author	Problem Addressed	Implementation and Results	Limitations/ Future Scope
Web Data Mining for Cyber Security Threat Detection By Tejal Bhimraj Ghuge, Sammip Sunil Biradar.	The rapid growth of online activity has led to more cyberattacks. This increase makes it harder to spot threats and weaknesses with traditional security methods.	The study used data mining techniques, including classification, regression, and association rule mining, to automate threat detection and predict future attacks. This approach led to better detection of anomalies, less manual effort, and improved identification of vulnerabilities.	The study used data mining methods, such as classification, regression, and association rule mining, to automate threat detection and predict future attacks. This approach resulted in better detection of anomalies, reduced manual effort, and improved identification of vulnerabilities.
The Performance Analysis of Cyber Threat Detection Model Using AI-based Classifiers By Navya Krishna Alapati, S.Dhanasekaran	Traditional cybersecurity tools, such as firewalls and antivirus systems, struggle to detect new threats. This leads to many false alerts and slow responses.	The study looked at several AI classifiers and deep learning models on labeled threat datasets. It reached higher accuracy, fewer false positives, better sensitivity, and improved ability to detect threats in real time.	The research shows that we need to handle big data better and improve our understanding of the field. It suggests that we can make future improvements by boosting collaboration between AI and humans and creating more scalable models.

2.7 Summary

The literature reviewed shows how artificial intelligence, machine learning, data-driven techniques are being rapidly integrated in finding solutions for modern cybersecurity threats. Article 2 emphasizes proactive, automated threat detection using AI-based layered systems with continuous learning, NLP, privacy-preserving methods to reduce false positives and improve response accuracy. Article 3 provides a comprehensive review of darknetenabled cybercrimes, emphasizes the development of adaptive and policy-driven defense mechanisms supported by advanced technologies such as AI, ML, quantum computing. Article 4 explores web data mining techniques, including, among others, association rule mining, regression, and classification, as effective approaches toward the prediction, detection, prevention of cyberattacks by uncovering hidden patterns in large datasets. Article 5 compares a number of AI-based classifiers and concludes that intelligent models significantly improve the detection accuracy while minimizing false alarms and enhancing scalability of defense systems.

When taken as a whole, the reviewed articles demonstrate that conventional reactive approaches are in fact inadequate to combat the threat landscapes of today. By enabling greater scalability, better predictive capabilities, and more intelligent security operations, AI-driven automation, when paired with data mining techniques and classifier-based models, offers a solid foundation for real-time cyber-threat detection and response.

Chapter 3

Software Requirements Specification

3.1 Functional requirements

- **User Authentication:** To restrict access to data, the system should allow safe login and registration for the designated users.
- **Uploading and Preprocessing datasets:** The system should allow users to upload datasets of network traffic and perform automatic preprocessing, including encoding, normalization, and handling of missing values.[5]
- **AutoML Model Training:** To automatically train and optimize the best-performing machine learning model for threat detection, the system should make use of an AutoML framework.
- **Cyber Threat Detection and Auto-Deletion:** The system should determine whether incoming traffic data contains high-risk or low-risk threats, and it should automatically remove or quarantine[3],[6] entries that are classified as such above a predetermined threshold.
- **Threat dashboard and Reports:** The system ought to provide users with a visual dashboard that shows the threats that have been identified, their severity levels,[5] and their timestamps.

3.2 Non-Functional requirements

- **Performance:** To ensure a smooth user experience, the system must generate threat detection results quickly.
- **Security:** All user data, login credentials, and threat logs must be safely kept in accordance with industry best practices.
- **Usability:** Even for non-technical users like analysts or faculty reviewers, the user interface must be straightforward, intuitive, and simple to use.
- **Reliability:** The system must function dependably in normal operating circumstances and continue to function correctly even in the face of unexpected inputs without crashing.
- **Scalability:** Without requiring major architectural changes, the system must be able to handle additional data or users in the future.

3.2.1 Safety Requirements

The AutoCTI system needs strong safety measures to securely handle network logs, real-time data, and automated threat responses. Key safety requirements include secure authentication and access control to prevent unauthorized use, encrypted communication channels to protect transmitted data, and privacy-preserving methods for storing or processing sensitive information.[2] Continuous monitoring and fail-safe controls are crucial to avoid accidental blocking or false automated actions[3],[2]. Moreover, the system must have audit logging, role-based permissions, and regular updates to guard against changing cyber attacks[7] and maintain safe operational performance in real-time environments.

3.2.2 Performance Requirements

- The system must process and classify cyber threat data (like network packets or log entries) in real time without crashing or experiencing memory overflow.

- It should be possible for mid-range CPUs and GPUs (such as an 8-core CPU or a 4GB GPU) to effectively identify and categorize threats.
- Inference latency should not exceed one second per input for real-time responsiveness.
- The system should achieve at least 90% classification accuracy on popular cybersecurity datasets.
- The model should continue to maintain precision and recall above 88% in order to guarantee low false positives and high detection coverage.

3.3 User interface design

The user interface of AUTOCTI is designed to ensure ease of use, clear visualization, and effective interaction for cybersecurity analysts. The key components include:

- **Log Upload Section:** allows users to send in system or network log files for offline analysis.
- **Live Monitoring Control:** Start/Stop button to initiate or halt real-time threat detection from live data streams.
- **Threat Visualization Panel:** Displays detected threats, threat level indicators, and affected system components.
- **Configuration Panel:** Provides options to set detection thresholds, enable automatic response actions, and configure notification settings.
- **Alerts Panel:** Lists detected threats with timestamps, severity levels, and suggested actions.

3.4 Hardware and Software requirements

Hardware Requirements

- Laptop/Dekstop

- CPU(intel core i7)
- RAM(16-32 GB)
- Storage: HardDisk,SSD
- GPU(optional -based on the complexity of the model)

Software Requirements

- **Operating System** - Windows 10/11
- **Programming Language** - Python with necessary libraries like BeautifulSoup, Requests,Pandas, NumPy etc
- **AutoML Platform** - Google Cloud AutoML (Natural Language / Tables)
- **Databases** - MongoDB
- **Cloud and API Services** - Google Cloud APIs and Threat Intelligence APIs
- **OS**- Development tools - VS Code,Jupyter Notebook / Google Colab

3.5 Any Other Requirements

- The project must include scripts for automatic dataset download, preprocessing, and splitting for training, validation, and testing.
- A benchmarking script should be included to compare the performance of different AutoML models (e.g., H2O AutoML, Auto-sklearn) across datasets like NSL-KDD, CICIDS2017, etc.
- The system should provide a detailed user manual or documentation for model training, evaluation, and deployment.
- The codebase must follow PEP8 Python coding standards and include clear and concise comments for maintainability and ease of understanding.

3.6 Summary

This explains the Software Requirements Specification for AUTOCTI, an AutoML-based automated cyber threat detection system. Secure user authentication, dataset upload, preprocessing, automatic machine learning model training, real-time threat detection with auto-removal, and dashboard reporting are among the system's functional requirements. Performance, security, usability, dependability, and scalability are among the non-functional requirements. In order to reduce false positives, the system must classify threats with low latency and operate efficiently on mid-range hardware while maintaining high precision and recall. Log upload, monitoring control, threat visualisation, configuration, and alerts are the modules that make up the user interface. To make it deployable and long-lasting, development will make use of Python, Google AutoML, MongoDB, documentation, benchmarking tools, and adhere to PEP8 standards.

Chapter 4

System Design

4.1 Abstract Design

4.1.1 Architectural diagram

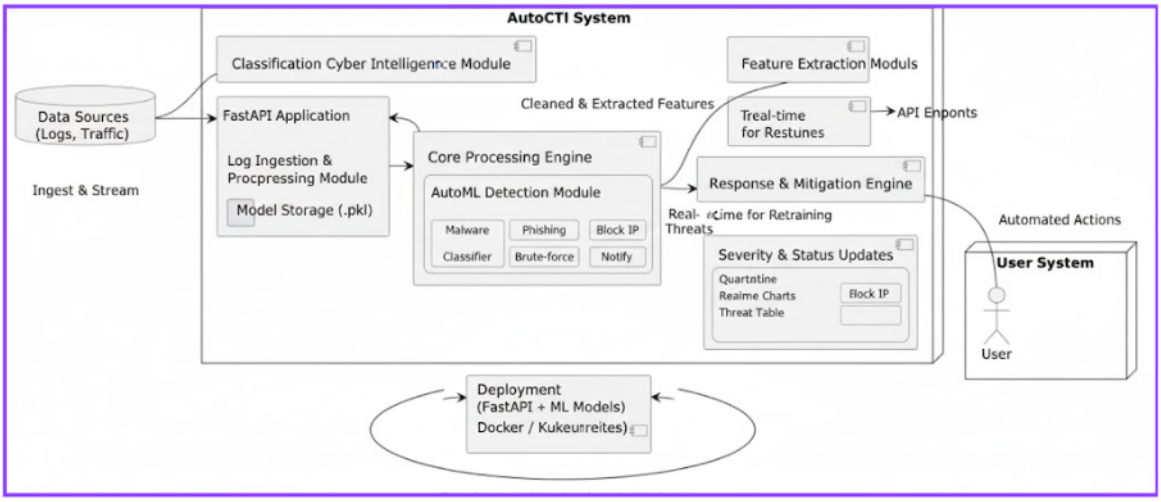


Figure 4.1: Architecture diagram

The diagram shows the structure of the AutoCTI system. It takes in log and traffic data, then processes and analyzes it for threat detection using machine learning. The main engine spots cyber threats and starts an automatic response, such as blocking or quarantining. It provides users with real-time updates and operates in a scalable environment using FastAPI and containerized models.

4.2 Proposed system

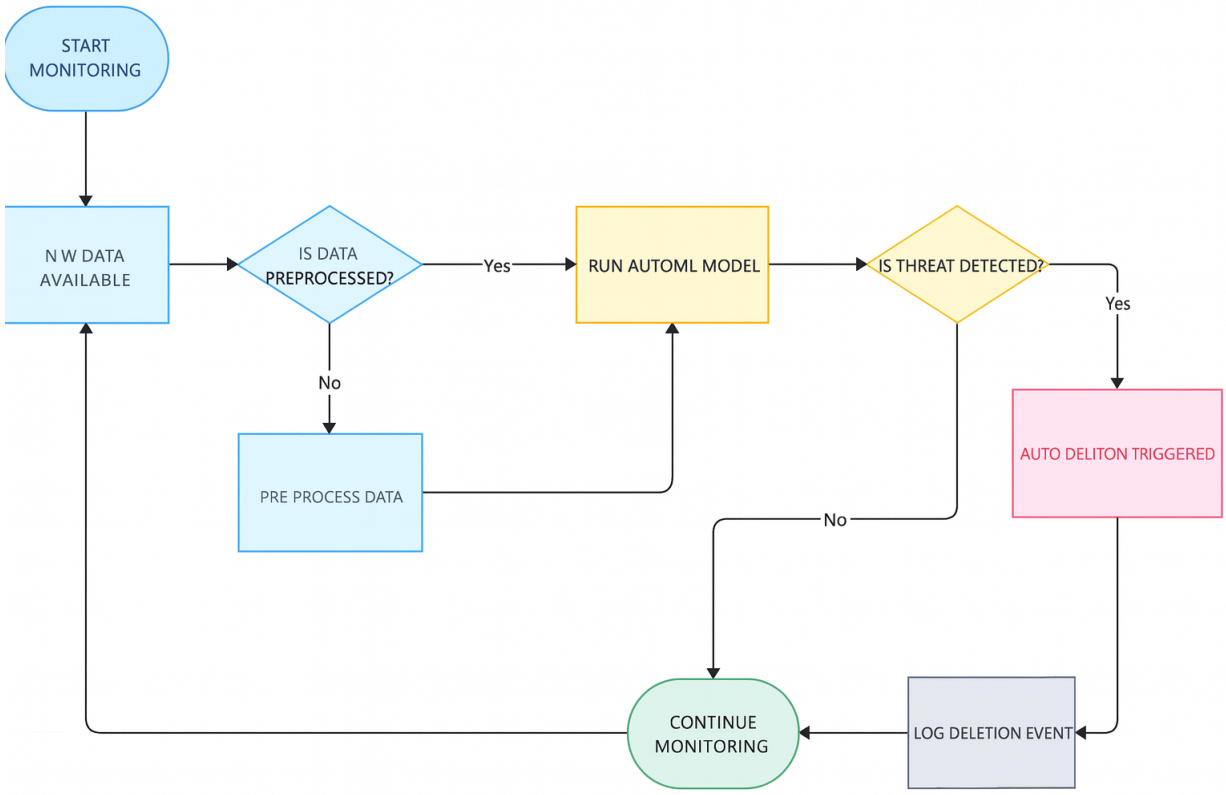


Figure 4.2: Proposed System

The first step in the suggested system workflow is to continuously monitor incoming data. The system verifies wheather newly available data has already undergone preprocessing. If not, the AutoML model is used to classify threats after the data has undergone preprocessing. To find out if there is a cyberthreat, the AutoML model examines the data. When a threat is found, the system logs the event for documentation and auditing purposes and initiates the deletion process to eliminate the threat. The system continues its monitoring cycle if no threat is found, guaranteeing continuous and real-time security surveillance.

4.3 Functional Design

4.3.1 Modular design diagram

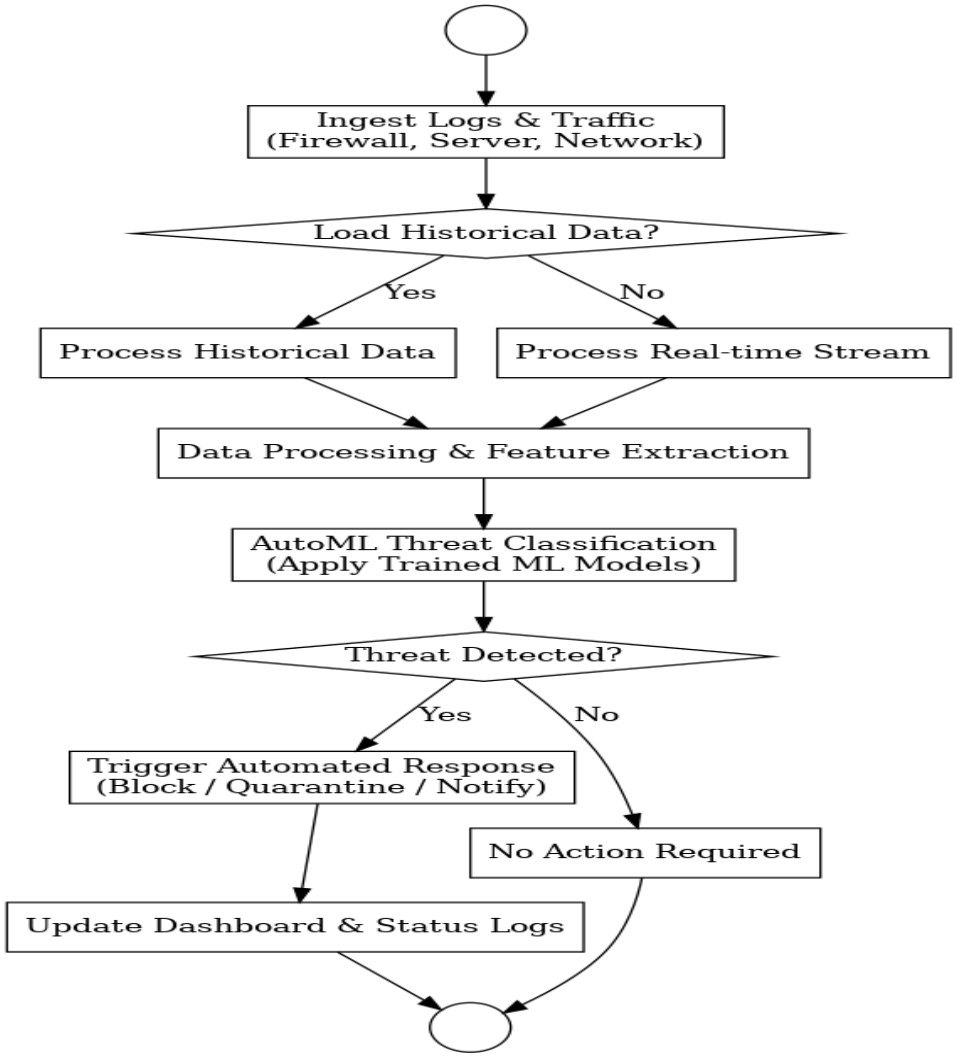


Figure 4.3: Modular design diagram

The modular diagram shows the workflow of the AutoCTI system. It starts with the collection of logs and network traffic. Then, it processes either historical or real-time data. Next, the system extracts features and classifies threats automatically using trained AutoML models. Depending on the classification result, it either triggers automated responses like blocking or notifications, or it takes no action if there is no threat. Finally, the module updates the dashboard and status logs to maintain ongoing monitoring and visibility of cyber threats.

4.3.2 Sequence diagram

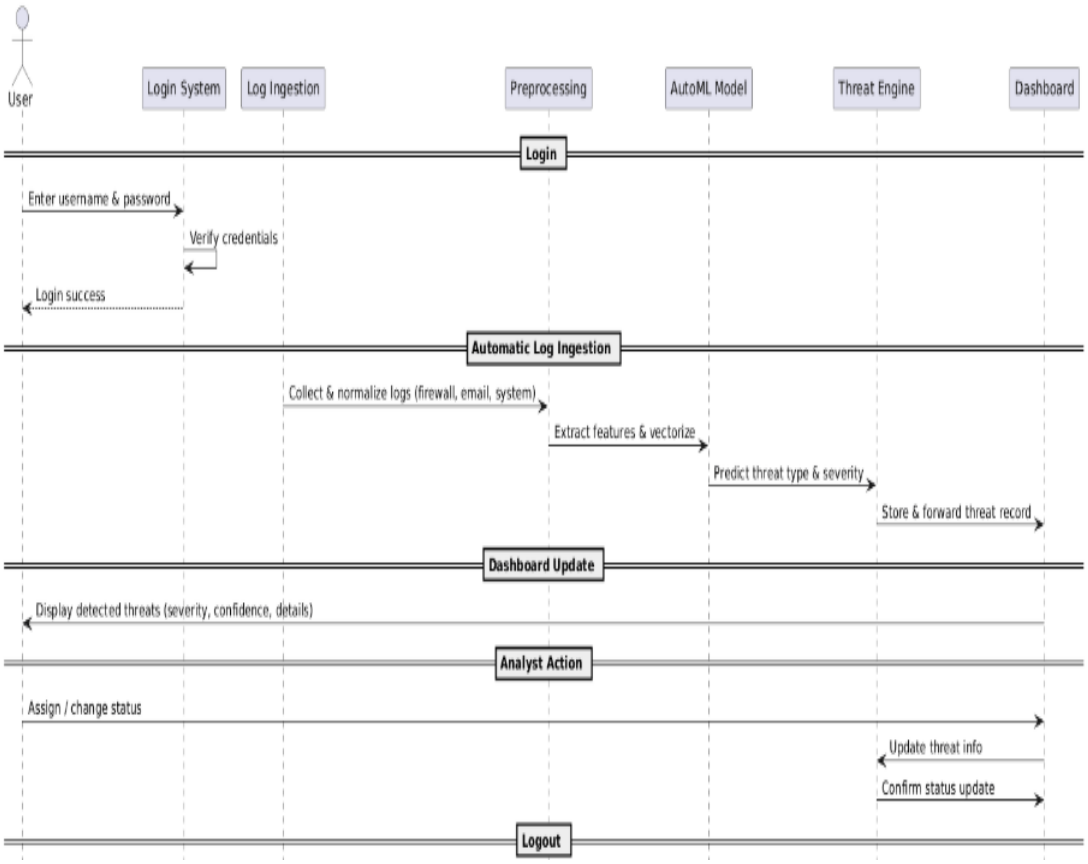


Figure 4.4: Sequence diagram

It describes the sequence diagram for the complete workflow of the AutoCTI system. This starts with user login, followed by automated log processing, threat prediction, displaying information on a dashboard, updating analyst status, and ending with successful logout. This shows how the system detects threats automatically while still allowing users to interact, respond, and monitor.

4.3.3 Use case diagram

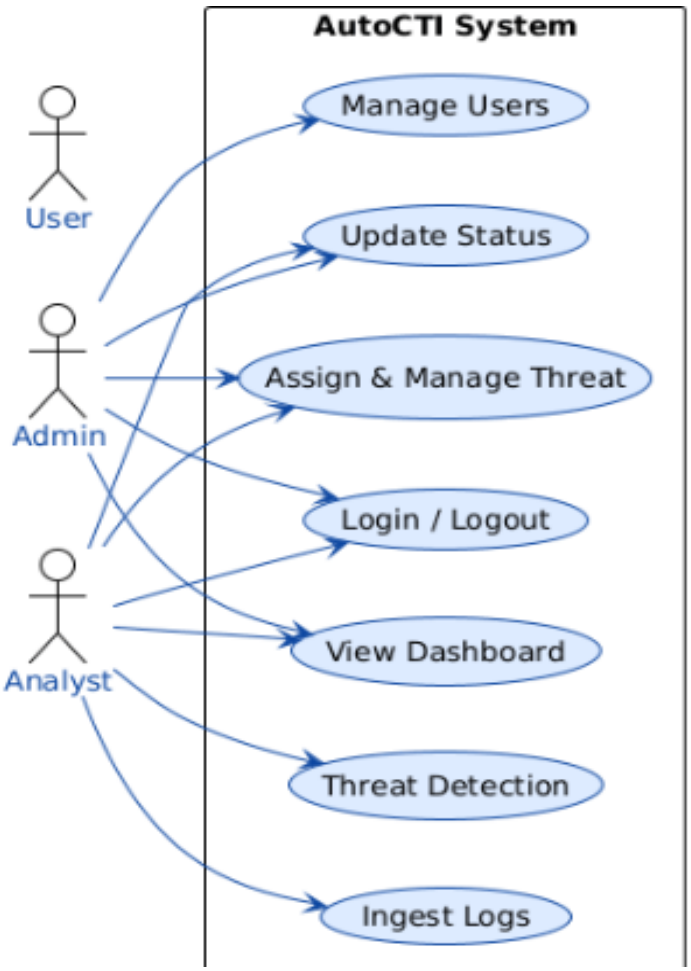


Figure 4.5: Use case diagram

The use case diagram shows the main functions of the AutoCTI system. It illustrates how users, admins, and analysts interact with the platform. The diagram highlights key actions such as log ingestion, threat detection, dashboard viewing, status updates, and user management. It demonstrates the system's role in supporting automated threat analysis while allowing for human oversight and administrative control.

4.4 Control Flow Design

4.4.1 Activity diagram for use cases

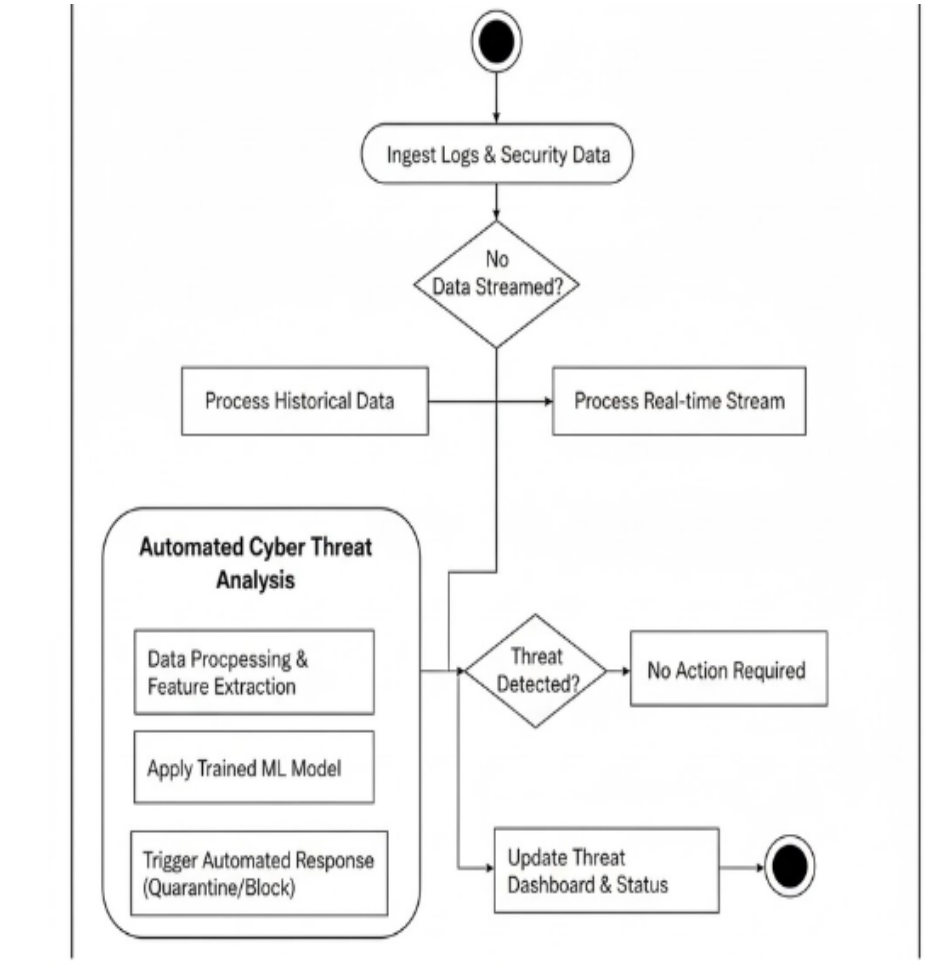


Figure 4.6: Activity diagram

The activity diagram shows the workflow for automated cyber threat analysis. It starts with the ingestion of log and security data. The data may be historical or real-time, and it gets processed and analyzed using a trained machine learning model. If the system detects a threat, it updates the dashboard and triggers automated responses like quarantine or blocking. If no threat is found, it takes no action. This flow highlights the system's capacity for continuous and automated threat monitoring.

4.5 Data Flow Diagram

4.5.1 Zero-Level Data Flow Diagram

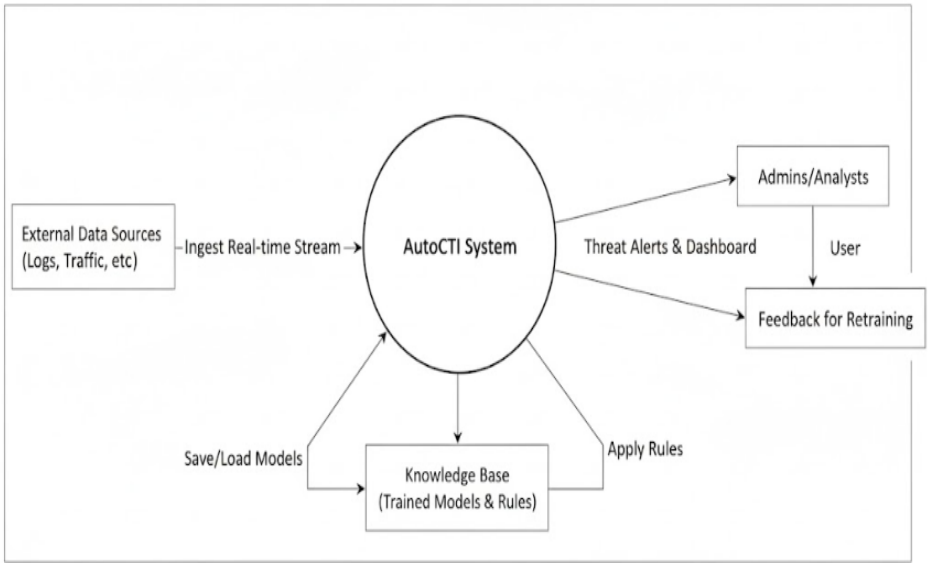


Figure 4.7: Zero-Level Data Flow Diagram

The diagram shows a zero-level view of the AutoCTI system. It ingests and analyzes real-time log and traffic data to create threat alerts and dashboard insights for admins and analysts. The system keeps a knowledge base of trained models and rules. This allows for automated decision-making and ongoing improvement from user feedback for retraining.

4.5.2 First-Level Data Flow Diagram

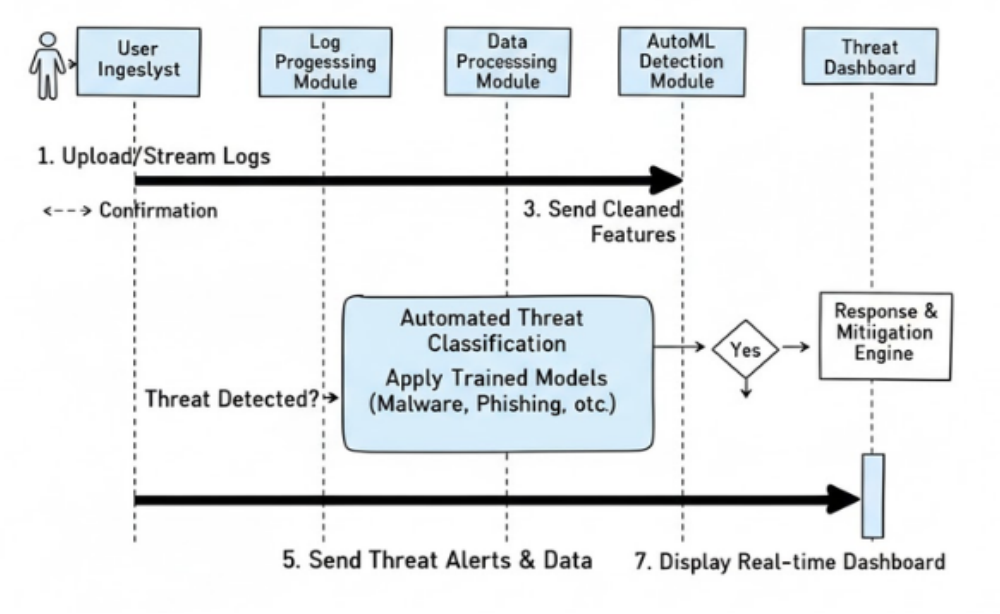


Figure 4.8: First-Level Data Flow Diagram

The diagram shows the first-level data flow of the AutoCTI system. Users upload or stream logs, which are processed and analyzed by data preprocessing and AutoML detection modules. If the system detects a threat, it sends alerts and forwards the information to the response engine. It also updates the real-time dashboard for monitoring. This flow illustrates the automated classification and reporting of cyber threats using trained machine learning models.

4.5.3 Second-Level Data Flow Diagram

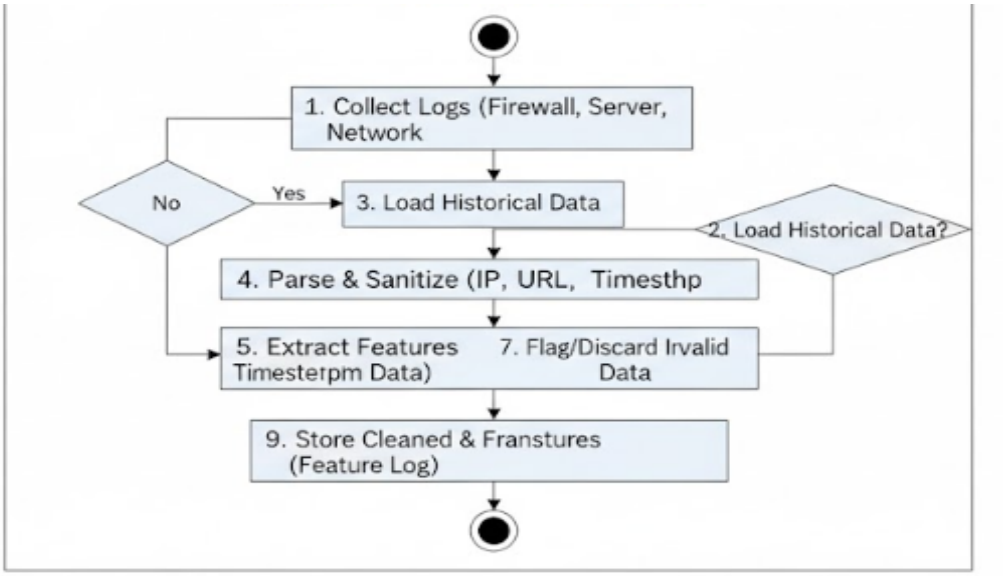


Figure 4.9: Second-Level Data Flow Diagram

The diagram illustrates the second-level process of log handling. Logs are collected from various sources and may be loaded from historical data. The system then parses and cleans details like IP addresses, URLs, and timestamps. It extracts key features and removes any invalid entries. Finally, the cleaned and organized data is stored for additional threat analysis.

4.5.4 Summary

This reveals the flow and design of the suggested AUTOCTI system. It starts with gathering current or past network logs, which are then automatically preprocessed, feature-extracted, and classified as threats using the models. Modular, sequence, and use case diagrams are then used to illustrate the functional design, displaying system features like user authentication, log processing, prediction, dashboard visualisation, and automated threat response. The control flow of how threats cause actions like blockage or an alert system while safe data continues to flow is shown by an activity diagram. The flow of data from collection to cleaning, storage, analysis, and visualisation is shown in data flow diagrams at the zero, first, and second levels. In conclusion, this chapter emphasises safe automation with human oversight and ongoing system enhancement.

Chapter 5

Implementation

5.1 Software used with justification

5.1.1 Frontend Development:

The AUTOCTI frontend is made to offer an easy-to-use interface for real-time threat activity monitoring. Users can upload datasets, see threats that have been identified, track severity levels, and keep an eye on automated system reactions like removing or isolating high-risk entries. Tables, alerts, and graphical elements are used in a responsive dashboard layout to guarantee that system outputs are clearly visualized. Threat status and logs are updated automatically without human intervention thanks to JavaScript's ability to provide real-time updates and seamless backend interaction through API calls. For both technical and non-technical users, the overall design prioritizes usability, clarity, and real-time accessibility.

5.1.2 Backend Development:

The AutoML model, automated threat response actions, and incoming data processing are all handled by AUTOCTI's backend. Preprocessing datasets, model training, severity classification, and the logic needed to isolate or remove identified high-risk threats are all handled by it.

Python is the primary backend language because of its robust support for automation and machine learning libraries. Through APIs, the backend and frontend exchange information, guaranteeing real-time updates

of system logs and detection results. Model outputs, alert history, and processed datasets are safely stored in MongoDB. In general, the backend guarantees secure handling of threat intelligence records, smooth model deployment, and dependable system operation.

5.1.3 Framework Used:

- FastAPI - Used to create the backend API that allows the model and frontend to communicate. It is appropriate for real-time threat detection and supports quick processing.
- Uvicorn - Utilized as the FastAPI application's ASGI server. It offers effective request handling and low latency performance.

5.1.4 Coding Languages used for Development :

- Python:

  Utilized for data preprocessing, model execution, backend development, and AutoML integration.

- JavaScript:

  Utilized in the front end for live threat visualization, API management, and dynamic content updates.

- HTML and CSS:

  Used to create the dashboard's user interface design and layout

5.1.5 Operating System:

Windows 10/11 is used to develop AUTOCTI because it offers compatibility with the necessary frameworks, libraries, and development tools. Additionally, it facilitates the smooth testing and execution of FastAPI, AutoML scripts, and the system's database services.

5.2 Hardware Used with justification

To guarantee seamless execution of data preprocessing, AutoML model training, and backend services, the system is built using a laptop with an Intel Core i7 processor and 16–32GB RAM. Faster read/write operations are made possible by SSD storage, which effectively stores datasets and system files. Although optional, a GPU is useful because it speeds up the model training process and enhances performance when working with big cybersecurity datasets.

5.3 Algorithm/procedures used in the project in different modules

- The raw cyber threat dataset is cleaned, encoded, and normalized by the preprocessing module to make it ready for training.
- Several machine learning models are automatically trained by the AutoML module, which then compares their performance metrics and accuracy to determine which model is best for deployment.
- Incoming data is classified by the prediction module using the trained model, which also determines the real-time threat severity level.
- Based on the threat's severity classification, the response module carries out automated tasks like storing, alerting, isolating, or deleting it.

5.4 Coding/pseudocodes

5.4.1 Introduction

The coding stage goal is to translate the system design into a working code that is compatible with the targeted programming languages and frameworks. Actually, separate tasks on different modules like preprocessing, AutoML model integration, prediction handling, and automated response

actions for maintaining the overall functionality of the AUTOCTI system have been completed here.

Various measures such as how readable, efficient, fast in execution, and maintainable the code is are the means through which the quality of the code written is judged. One of the main goals of programming is to have a clear and well-organized code and it serves as a way of lessening the debugging, updating in the future, and scaling the system as well. In addition, following coding standards and writing documentation is not only a good maintenance habit for the distant future but it also facilitates the next stages of development in the project.

5.4.2 Programming Style

In AUTOCTI, the developers have kept up a uniform programming style throughout which different modules' functional flow is not only made easy but also quite noticeable by the implementation's clarity, extendibility, and ease of maintenance. As the project is a combination of data preprocessing, AutoML execution, prediction handling, and automated threat response, a carefully designed coding strategy is very helpful in steering clear of mistakes and making it easier to update or change the system in the future.

- Data Preprocessing Module: The module takes charge of refining the raw dataset about cyber threats that is to be used for the subsequent training. It is a part of the pipeline that features operations such as deletion of unused records, treatment of missing values, conversion of text-based fields, and normalization of data formats. When the preprocessing is done properly, the AutoML engine gets a clean and well-structured input, which makes the model accuracy higher and more stable during the run.

- AutoML Model Training Module: Different machine learning models can be automatically trained and then evaluated by various performance metrics such as accuracy, precision, recall, and F1-score through the use of this module. Therefore, there is absolutely no

need for a manual model selection and hyperparameter tuning. The system is the one that decides the best performing model, and hence, it is the one that is saved for deployment, which, at least partially, makes the whole detection process reliable and efficient.

- Threat Classification Module: This module utilizes a trained model to examine the threat information given and to determine the threat level. Since the classification is carried out quickly, it can be utilized for the alert-based response. To keep the correctness of the decisions, the module changes the model prediction results to the local severity categories (low, medium, and high) of the already existing threats.

- Response Automation Module: According to the categorization result, the apparatus performs the required changes on its own.Malware, to give an example, can be removed or separated immediately while a less harmful kind of file may just be noted in a log file for later checking. The proportion of manual work that is reduced due to the automation of these tasks, the time of the reaction that is shortened, and the overall level of the system's efficiency in handling threats that is also increased are all effects that result from this.

- Dashboard Module: This module provides an enjoyable manner to comprehend the system and see the detection results. In fact, it is very simple and intuitive in displaying the detected objects, their location, time, and the actions that were executed automatically. Moreover, the security analysts by means of the dashboard, are able to keep a check on the system's performance and have a glance at the processed results without the need of accessing the backend code directly.

5.5 Summary

The software implementation described here was the product of AUTOCTI, which also encompassed not only the central system but the different modules and the programming style that were uniformly propagated throughout the system. Each module like preprocessing, AutoML model training, threat classification, automated response handling, and dashboard display was designed to carry out a specific function in the cyber threat detection process be it accurate and efficient. The entire system was implemented with a consistent and well-organized programming approach which made it readable, scalable, and upgrader or changable at a later time without much effort.

Chapter 6

Results and Discussion

6.1 Results

This Section includes the presentation of results using suitable figures, graphs and charts with proper Explanations.

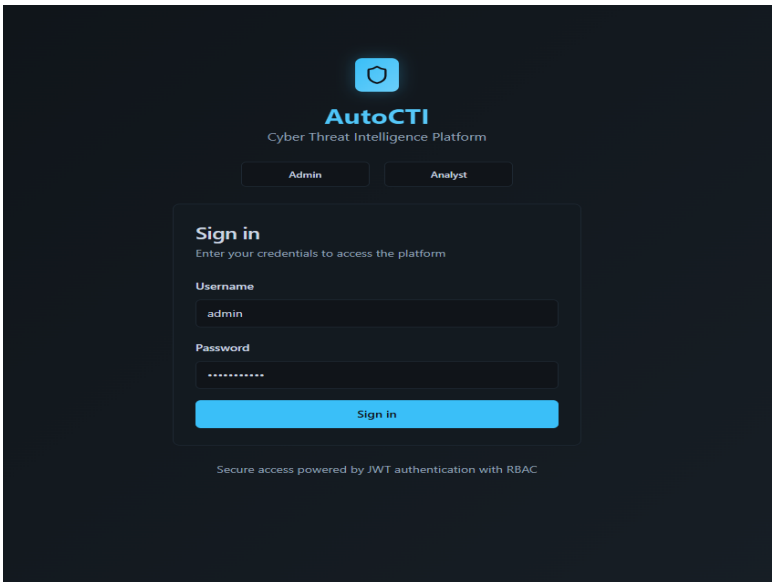


Figure 6.1: Result-1 (Login page)

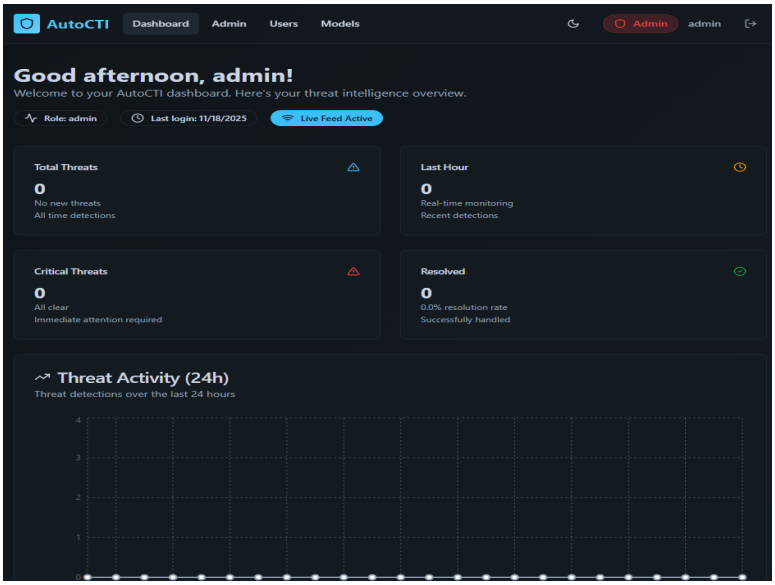


Figure 6.2: Sample Result-2 (Dashboard)

6.2 TESTING:

The software testing focused primarily on the aspects of correctness, security, and completeness of the AUTOCTI system, as well as the quality of the system overall. Tests are basically scenarios in which the system is run with some predefined inputs, and their objective is to locate defects or deviations from the expected behavior. The main objective is to guarantee that the system is able to perform accurate cyber threat detection, classification, and automated response handling, as it has been programmed already.

When testing a system, it is taken through an assortment of predefined tests, and the real outcomes are contrasted with the anticipated ones. The most important factor that, however, overwhelmingly decides the effectiveness of the tests is the quality and coverage of the test cases. The test cases in AUTOCTI are targeted at verifying the accuracy of the prediction model, a request's efficient performance, data manipulation, and dashboard reporting.

A test case is an extensive testing file which defines the input, the processing, the expected output, and the actual result. Moreover, it may

detail certain situations or examples, such as the employment of high-risk inputs, the validation of the dataset, or the prediction being initiated by some operations.Every test case is a facility to record the output obtained that can be matched with the expected output for checking whether the system works properly.

6.3 TYPES OF SOFTWARE TESTING:

- Black Box Testing: Black box testing refers to the method in which the internal structure or code of the AUTOCTI system is basically not known. Firstly, input datasets and test cases are executed, and the output is later checked for the correct threat detection, severity classification, and response actions. This way the system's functionality can be confirmed from the viewpoint of a user.
- White box testing: White box tests are mainly aimed at checking the internal logical flow and code execution of AUTOCTI. Besides, they also perform checks on several other operations, like the verification of preprocessing functions, AutoML model handling, prediction logic, and automated response flow. Therefore, it is essentially a detailed operation to ensure that the internal implementation is not only correct but also efficient.

6.4 TESTING METHODOLOGY

The different types of testing are as follows:

6.4.1 Unit Testing:

Individual modules of the AUTOCTI system were unit tested to make sure that each component functions correctly by itself before the components are integrated. Modules such as data preprocessing, AutoML model training, threat classification, and automated response handling were all tested

individually with both valid and invalid inputs. This allowed for the detection of logical errors at the very first stage and also made it possible to verify that each module produces the expected output. Unit testing was the way in which the core functions of the system were made reliable and became suitable for the next layer of integration and system-level testing.

6.4.2 Integration Testing

Integration testing was carried out after unit testing was successfully completed in order to confirm the integration of all AUTOCTI modules into one system. The modules such as preprocessing, AutoML model execution, prediction, automated response, database storage, and dashboard interface were combined and tested for the proper flow of data and communication. The tests have uncovered issues that were interactions between modules, API communication, and response triggering, thus making sure that the system is precise and stable in the situations when the components are working together.

6.4.3 System Testing:

System testing through various stages of operations was done after the integration of all the AUTOCTI system modules to check the overall functionality and performance. The final system was exercised with both actual and simulated cyber threat data to check if all the operations of data preprocessing, model prediction, automated response, and dashboard display were as expected. The test has verified that the system meets the specifications and is able to make correct threat detection, severity classification, and hence, carry out the automated actions in different scenarios.

6.4.4 Acceptance Testing:

Acceptance testing was performed to verify that the AUTOCTI system meets the necessary specifications and performs all the functions as per the project scope. Different features of the system such as ease of use,

correctness of predictions, severity classification, automation of response actions, and the dashboard functionality were considered for the assessment. The examinations confirmed that the system behaves as expected from the end-user perspective and, therefore, it may be deemed suitable for release in the real world.

6.5 TESTING CRITERIA

6.5.1 Testing for Threat Detection

Table 6.1: TESTING CRITERIA for Threat Detection

Test Case	Input	Test Description	Output
1	Known malicious threat data	To check if the system correctly detects a malicious input	System identifies the input as a threat
2	Clean/safe data	To verify that no false threat is detected	System shows no threat detected

6.5.2 Testing for Severity Classification

Table 6.2: TESTING CRITERIA Severity Classification

Test Case	Input	Test Description	Output
1	High-risk malicious sample	To verify if the system assigns the correct severity level	System classifies the input as High Severity
2	Suspicious or partially harmful data	To check if the system assigns a mid-level severity	System classifies the input as Medium Severity
3	Low-risk or harmless dataset	To ensure no incorrect high-severity tag is applied	System classifies the input as Low or No Severity

6.5.3 Testing for Automated Response

Table 6.3: TESTING CRITERIA for Automated Response

Test Case	Input	Test Description	Output
1	High-severity threat	To check if the system performs automatic deletion/isolation	System triggers automatic delete or isolate action
2	Low or no threat input	To ensure no unnecessary action is triggered	System performs no response and simply records or ignores input

6.5.4 Testing for User Interaction

Table 6.4: TESTING CRITERIA forUser Interaction

Test Case	Input	Test Description	Output
1	User submits dataset through dashboard	To verify if the system accepts input and responds without delay	Dataset successfully uploaded and acknowledged
2	User views prediction results	To check if threat details, severity, and responses are displayed properly	System displays clear and readable results

6.6 Comparative Analysis

Table 6.5: Comparision of Existing Work and Results Obtatined

Parameters	**Existing System**	**Proposed System**
Threat Detection	Mostly manual review and rule-based filtering	Automated threat detection using AutoML
Severity Classification	Not available or done manually	Automatically classifies threats as Low, Medium, or High severity
Response Handling	Requires security team intervention	AAutomatic response execution such as deletion, isolation, or alert
Model Adaptability	Fixed rules that require manual update	Learns and improves with new threat data
Accuracy	Moderate and depends on analyst experience	Higher accuracy due to trained ML model and preprocessing
User Accessibility	Requires cybersecurity expertise to operate	User-friendly dashboard suitable for both technical and non-technical users
Scalability	Limited and difficult to extend	Highly scalable due to modular architecture and automated learning
Overall Efficiency	50–65% depending on user involvement	85–95% based on automated processing and model performance

6.7 Summaary

The AUTOCTI testing process was a confirmation of the new system whether it complied with the established requirements and operated as expected. Various tests such as unit, integration, system, and acceptance tests were conducted to verify functionality, correctness, automated response behaviour, and system performance. Test scenarios were derived from concepts such as threat detection, severity classification, automated response execution, prediction speed, and user interaction. As per the findings, the system accurately detects cyber threats, extremely fast determines their severity, and therefore, is capable of initiating the appropriate automated actions.

Chapter 7

Conclusions and Future work

The main goal of the AUTOCTI program was to implement AutoML in a way that would lead to the automatic and therefore fast identification of cyber threats, thus resulting in risk identification that is not only faster but also more accurate. During different phases of the initiative, the functionalities of the system, like data preprocessing, AutoML model training, classification, automated response, and dashboard visualization, were progressively integrated to achieve operational efficiency. The device is able to receive data on cyber threats, evaluate the aggressiveness of the threats, and if the risk is high, it can thus decide to record, separate, or delete the entries without a human intervention.

The AUTOCTI system is capable of various enhancements to not only raise its operational level but also to make it more adaptable to the cybersecurity scenarios taken from the real world. It is quite probable that at a later time, the system will be broadened to incorporate data streams in real-time from SIEM tools and network monitoring live feeds. Moreover, the AutoML model can be transformed to have a higher precision and fewer false predictions through the use of advanced methods like deep learning. In addition, the system having access to threat confirmation by being connected to external threat intelligence APIs may lead to the quickness of the decision-making process. Furthermore, the system can be evolved to incorporate multi-user access with roles, big-scale deployment in the cloud, and automated mitigation, for instance, blocking malicious IPs or triggering incident response workflows, etc., rather than simply deletion.

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DOI : https://doi.org/10.1109/GCCIT63234.2024.10862442.

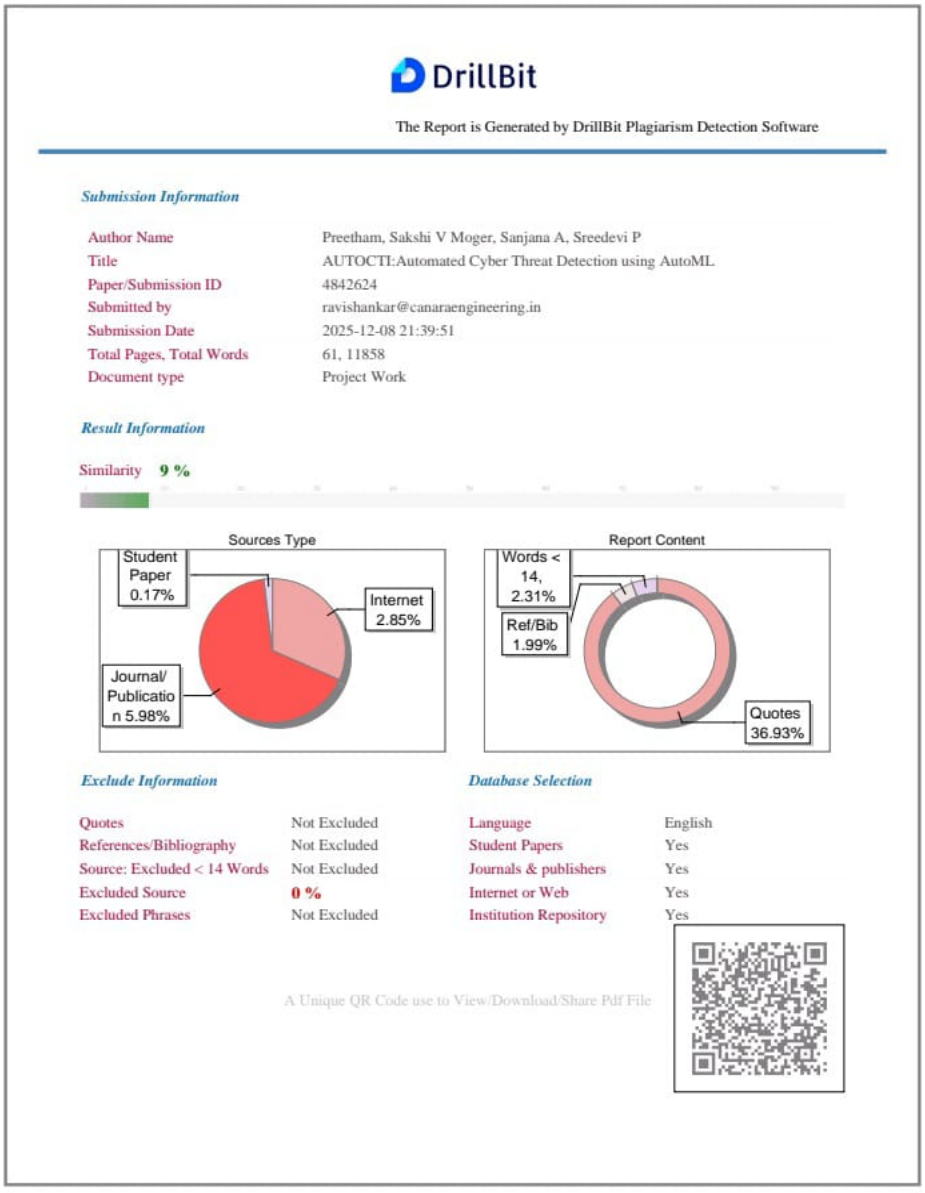
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DOI :https://doi.org/10.1109/ACCESS.2024.3493957.

Appendix A

Drill-bit Plagiarism Report



DrillBit

The Report is Generated by DrillBit Plagiarism Detection Software

Submission Information

Author Name	Preetham, Sakshi V Moger, Sanjana A, Sreedevi P
Title	AUTOCTI:Automated Cyber Threat Detection using AutoML
Paper/Submission ID	4842624
Submitted by	ravishankar@canaraengineering.in
Submission Date	2025-12-08 21:39:51
Total Pages, Total Words	61, 11858
Document type	Project Work

Result Information

Similarity 9 %

Sources Type

Student Paper 0.17%

Internet 2.85%

Journal/ Publicatio n 5.98%

Report Content

Words < 14, 2.31%

Ref/Bib 1.99%

Quotes 36.93%

Exclude Information

Quotes	Not Excluded
References/Bibliography	Not Excluded
Source: Excluded < 14 Words	Not Excluded
Excluded Source	0 %
Excluded Phrases	Not Excluded

Database Selection

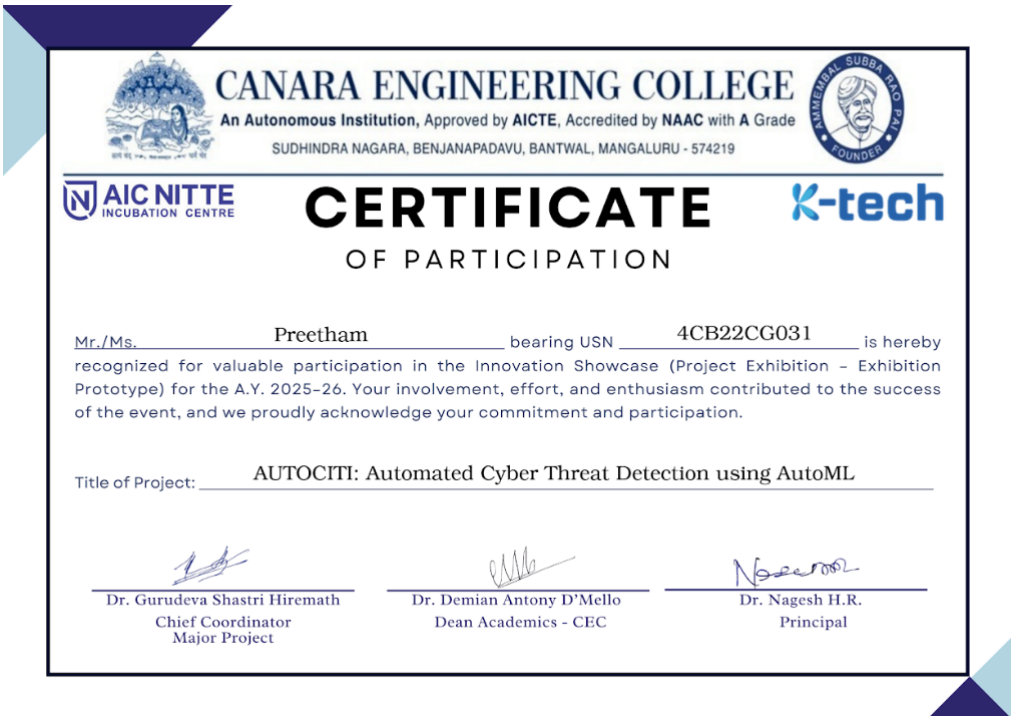
Language	English
Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
Institution Repository	Yes

A Unique QR Code use to View/Download/Share Pdf File

Figure A.1: Drill-bit Plagiarism Report

Appendix B

Project Expo Details



CANARA ENGINEERING COLLEGE

An Autonomous Institution, Approved by AICTE, Accredited by NAAC with A Grade

SUDHINDRA NAGARA, BENJANAPADAVU, BANTWAL, MANGALURU - 574219

AIC NITTE INCUBATION CENTRE

CERTIFICATE

OF PARTICIPATION

K-tech

Mr./Ms. Preetham bearing USN 4CB22CG031 is hereby recognized for valuable participation in the Innovation Showcase (Project Exhibition – Exhibition Prototype) for the A.Y. 2025-26. Your involvement, effort, and enthusiasm contributed to the success of the event, and we proudly acknowledge your commitment and participation.

Title of Project: AUTOCITI: Automated Cyber Threat Detection using AutoML

Dr. Gurudeva Shastri Hiremath
Chief Coordinator
Major Project

Dr. Demian Antony D'Mello
Dean Academics - CEC

Dr. Nagesh H.R.
Principal

Figure B.1: Preetham - 4CB22CG031